Patent Infringement Analysis: US 9,354,742 B2 — "Foldable Electronic Device and Method of Managing Visible Regions Thereof" (Samsung / Samudrala Nagaraju)

TL;DR

- The patent's independent claims are most likely INFRINGED on a literal-claim-construction basis by outward-folding wraparound foldables — chiefly the **Huawei Mate X / Mate Xs / Mate Xs 2**, **Huawei Mate XT** tri-fold, and the **Royole FlexPai 1 / FlexPai 2** — because those devices use a single continuous flexible display whose "rear" portion runs an independent application from the front "flat" portion when folded, mirroring the patent's "rear display unit" / "curved visible region" language almost verbatim.
- For book-style foldables with a SEPARATE cover screen (Samsung Galaxy Z Fold/Flip, Google Pixel Fold, OnePlus Open / Oppo Find N3, Honor Magic V3 / V5, Xiaomi Mix Fold / Flip, Motorola Razr Plus / Fold), literal infringement is materially weaker because the patent specifies a "rear display unit" of a single foldable body and consistently refers to a "curved visible region" generated by bending; doctrine-of-equivalents arguments remain plausible but contestable.
- Samsung owns the patent and so cannot infringe its own claims; no IPR, ITC, or district-court action asserting US 9,354,742 B2 has been identified, and the recent **Lepton Computing LLC v. Samsung** suit (Case No. 2:26-cv-00338, E.D. Tex., filed April 23, 2026 per PacerMonitor/Justia docket records) involves nine different Lepton-owned patents (oldest filed June 29, 2021) and does not implicate this patent. WebProNews + 2

Key Findings

1. Patent scope and unusual language. Independent Claim 1 (method) and Claim 10 (device) require (a) a foldable device with both a **front display unit** AND a **rear display unit**, (b) detection of bending, (c) determination of a **flat view area** and a **bent view area**, and (d) provision of a UI in the bent view area of the **rear display unit** that is associated with applications **different from** the application shown in the flat view area. Dependent claims add touch sensing in both areas, angle-threshold detection, single-application UI split across both areas, app re-layout on unbending, viewport resizing on squeeze, and content scrolling on roll. The 2014 specification — quoted verbatim in Patently Mobile's coverage of the published Samsung application — speaks of a "curved visible region" and "flat visible region" generated by "folding and/or rolling" a single continuous foldable display, framing the invention squarely in the wraparound-OLED paradigm Samsung Display was prototyping at the time, NOT the book-style hinged dual-screen paradigm that dominated commercialization after 2019. Patently Mobile

2. Strongest literal-infringement candidates: Huawei outward-folding Mate X family and Royole FlexPai. Both employ a single bent-around flexible OLED in which the "front" and "rear" halves are physically two regions of one continuous panel — the cleanest fit for the patent's "front display unit" / "rear display unit" wording and its "curved visible region" embodiment.

Royole's FlexPai 2 (Sept 22, 2020) demonstrably "each screen behaves like its own homescreen, and each screen can run its own app" (XDA Developers), satisfying the "different applications" limitation of Claim 1. XDA Developers

3. Book-style foldables present a tougher literal read but plausible DOE exposure. Samsung Galaxy Z Fold/Flip, Google Pixel Fold, OnePlus Open, Honor Magic V3, Xiaomi Mix Fold/Flip, and Motorola Razr Plus/Fold all detect a folded state via Hall-effect sensors, partition UI between a main inner display and a separate outer cover/Flex Window, and routinely run different apps on the two screens. Under literal construction, the "rear display unit" of Claim 10 maps imperfectly to a physically separate cover panel; under doctrine of equivalents a fact-finder could find the substitution insubstantial because the function (showing distinct content when folded), way (sensor-triggered partition), and result (two independent UI areas) are arguably the same.

4. Flex Mode / Hover Mode / Tent Mode partial-folding features are tangential to Claims 1 and 10 because those claims require distinct content on the FRONT and REAR display units, not two halves of the same inner display. Flex Mode maps more cleanly to dependent claim 6 only under an expansive construction.

5. No identified litigation or IPR on this specific patent. A targeted PTAB / PACER / Bloomberg Law search returned zero proceedings asserting or challenging US 9,354,742 B2.

Details

Patent Claim Breakdown Matrix

Element	Required Feature
Claim 1(a)	Foldable device with **front display unit AND rear display unit**
Claim 1(b)	Detect if device is bent
Claim 1(c)	Determine **flat view area** AND **bent view area** in response to bending
Claim 1(d)	UI in **bent view area of rear display unit** with apps **different from** app in flat view area
Claim 2	Touch gesture sensing on flat or bent area
Claim 3	Relative-angle threshold gate for bend detection
Claim 4	Single application UI split across both regions
Claim 5	Application re-layout on bent → unbent transition
Claim 6	UI displayed on BOTH flat (front) AND bent (rear) areas
Claim 7	Viewport resize on squeeze event
Claims 8–9	Scroll on roll event

Element	Required Feature
Claim 10 (device)	Hardware mirror of Claim 1 — touch-sensitive body, microprocessor configured as above
Claims 11–16	Dependent device features mirroring 2–9

Product-by-Product Analysis

A. Huawei Mate X / Mate Xs / Mate Xs 2 (outward-folding, single wraparound OLED)

- **Launch:** Mate X — November 15, 2019 in China at 16,999 yuan (~$2,403), per CNBC (Oct 23, 2019). Mate Xs — global March 2020; Mate Xs 2 — May 2022. (CNBC) (Wikipedia)
- **Display configuration:** Single 8-inch flexible OLED folds outward; when folded, presents a 6.6-inch "daily driver" front panel and a 6.38-inch rear panel — both portions of the SAME continuous display (Wikipedia: "The Mate X has an 8-inch OLED display that can fold outwards resulting in a 6.6-inch main display and a 6.38-inch rear display"). PCWorld confirmed the rear panel can act as a mirrored selfie viewfinder, demonstrating different UI on the rear region than the front. (Wikipedia) (PCWorld)
- **Bend detection:** Falcon-wing hinge; folded vs unfolded states drive UI partition.
- **Software:** EMUI partitions the single panel into independent front and rear regions; rear region can show a different application (e.g., selfie viewfinder while main camera UI is on front).
- **Claim-by-claim:**
  - 1(a) **Met literally** — front and rear regions of one wraparound display are functionally "front display unit" and "rear display unit."
  - 1(b)–(c) **Met** — hinge sensor detects fold; UI partitions into flat front and bent rear regions.
  - 1(d) **Met** — rear shows different UI (e.g., selfie viewfinder vs camera control) in built-in apps.
  - Dep. 2, 6 **Met** — touch active on both regions, UI displayed on both.
  - Dep. 5 **Met** — app re-layout on unfold to full 8-inch.
  - Dep. 3 likely met via hinge-angle threshold; dep. 7–9 (squeeze/roll) **not met**.
- **Severity: HIGH** — Mate X is the closest commercial embodiment to the patent's described invention.
- **Confidence: High** for literal infringement of independent claims.
- **Design-around difficulty:** Very difficult — the hardware itself directly embodies the claim limitations.

B. Royole FlexPai 1 and FlexPai 2 (single outward-folding wraparound OLED)

- **Launch:** FlexPai 1 announced October 31, 2018 in Beijing; order fulfillment began "late December, 2018," per Royole's official press release (Nov 2, 2018, via Yole Group /

Tweaktown). FlexPai 2 launched September 22, 2020 at 9,988 yuan (~$1,470). (Grokipedia + 2)

- **Display:** 7.8-inch flexible OLED, outward-folding wraparound. When folded, FlexPai 1 presents two ~4-inch outward-facing halves (16:9 and 18:9) plus a 21:6 hinge "spine" strip; FlexPai 2 presents a 5.5-inch front and 5.4-inch rear half. (Android Authority + 3)
- **Software:** Royole's waterOS 2.0 (Android 10 fork) treats each folded half as an independent homescreen. XDA Developers (FlexPai 2 preview): *"each screen behaves like its own homescreen, and each screen can run its own app. For example, I can have Instagram running on the front screen, and WhatsApp running on the back screen."* Engadget confirms hinge-strip shortcuts. (Engadget) (XDA Developers)
- **Claim-by-claim:** All elements of Claim 1 and Claim 10 met literally, including the "different applications" limitation. Dependents 2 (touch), 3 (angle), 5 (re-layout), 6 (UI on both sides) met. 7–9 (squeeze/roll) not implemented.
- **Severity: HIGH. Confidence: Very High** for independent claims.
- **Note:** Royole declared bankruptcy in 2024 — The Shenzhen Intermediate People's Court officially accepted the bankruptcy case on May 15, 2024 (CTOL Digital Solutions, citing Chinese court records); bankruptcy liquidation proceedings began June 12, 2024 (Digitimes). Commercial relevance is therefore limited, but pre-2024 sales remain actionable within the 6-year statutory damages window. (CTOL Digital Solutions) (DIGITIMES)

C. Huawei Mate XT Ultimate Design (tri-fold, Sept 10, 2024)

- **Display:** 10.2-inch flexible OLED with two hinges (one inward, one outward; Z-fold). Folds to 6.4-inch single-screen mode, 7.9-inch dual mode, or 10.2-inch tri-screen mode. (TechInsights) (Android Authority)
- **In dual or tri-fold state, sections show different apps:** Designboom: *"For dual mode, they can open another screen and have different apps displayed on each."* (Designboom)
- **Claim mapping:** The outward-folding portion produces a literal "rear" face of the same flexible display. When in dual-screen mode, the rear-facing section runs different apps from the forward-facing section — meeting Claim 1. Claim 10's device limitations (touch-sensitive flexible body, microprocessor, hinge sensors) also met.
- **Severity: HIGH. Confidence: High** for claims 1, 10; Medium for dependents (Mate XT lacks Flex Mode per SamMobile).

D. Samsung Galaxy Z Fold series (Fold–Fold 7, Fold SE, Z TriFold) and Z Flip series (Flip–Flip 7)

- **Self-implementation; cannot infringe** (Samsung owns the patent).
- Included only for claim-scope calibration: Samsung's "Continue apps on cover screen" feature and Flex Mode are clear commercial embodiments of Claim 1's bent/flat partition concept. Samsung Developer documentation states for Flex Mode: *"In Flex Mode, the display is divided into two parts… Content is shown on the top screen and the content title and controls are shown on the bottom screen."* This corresponds to dependent claim 4 (single app split across two areas) and claim 6 (UI on both). (Samsung Developer)

- **Use as benchmark:** Demonstrates that Samsung itself interprets its foldable patent portfolio broadly enough to cover separate cover-screen and main-screen UIs.

E. Google Pixel Fold / Pixel 9 Pro Fold / Pixel 10 Pro Fold (book-style)

- **Display:** Separate inner foldable display and outer cover display; Hall-effect / hinge sensors detect fold.
- **App Continuity:** Limited at launch (video and Google Maps only). Per 9to5Google (Abner Li, Sept 20, 2023): *"Besides letting you force aspect ratios, Android 14 QPR1 adds a 'Continue using apps on fold' setting on the Pixel Fold."* Android 15 added a swipe-up-to-continue gesture similar to OnePlus Open (Android Authority, Android Headlines). Android Headlines + 3
- **Claim mapping:** Literal "rear display unit" reading is contestable — the cover screen is physically separate, not a "rear" region of one display body. Under DOE: cover and main display show different apps in many scenarios → arguably equivalent.
- **Severity: MEDIUM (DOE) / LOW (literal). Confidence: Medium.**

F. OnePlus Open / Oppo Find N / Find N2 / Find N3 / Find N5

- **Display:** Separate cover + inner foldable; Hall-effect hinge sensor.
- **App Continuity:** OnePlus pioneered the "swipe-up-to-continue" gesture later cited by Google as a model; allows cover screen to keep running the previously inner-displayed app or run a different one. Android Authority PhoneArena
- **Claim mapping:** Same analysis as Pixel Fold — separate cover screen weakens literal infringement; DOE plausible.
- **Severity: MEDIUM (DOE) / LOW (literal). Confidence: Medium.**

G. Honor Magic V / V2 / V3 / V5 / Vs / Vs Ultimate, Magic V Flip / V Flip2

- **Display:** Separate cover and inner foldable. MagicOS 8/10 with extensive cross-screen workflows including Face-to-Face Translation: the main inner display shows your text while the cover screen simultaneously shows the translated text to the other party — two independent UI streams on the front (cover) and back (main) faces of the folded device, physically reminiscent of the patent's "front display unit" / "rear display unit" pair. Amateur Photographer
- **Severity: MEDIUM (DOE), borderline HIGH for the Face-to-Face Translation use case. Confidence: Medium.**

H. Xiaomi Mix Fold 1/2/3/4 and Mix Flip

- **Display:** Separate cover and inner foldable; Mix Fold 4 (2024) supports "Hover Mode" (Samsung Flex Mode analog) and app continuity. GSMArena Mix Fold 3 review describes three "After folding" options: *"conditionally continue the app on the cover screen if you* *swipe up within 6 seconds, continue the app without the need to swipe, or simply lock the phone."* GSMArena

- **Severity: MEDIUM (DOE) / LOW (literal). Confidence: Medium.**

I. Motorola Razr / Razr Plus / Razr 40 Ultra / Razr 50 Ultra / Razr Fold

- **Display:** Clamshell foldable inner + large external cover (3.6"–4" on Razr Plus/Ultra; 6.6" on Razr Fold). MakeUseOf
- **Software:** Motorola runs FULL Android apps on the cover screen; "Continue apps on cover screen" via Settings → External display → App settings (Android Central, Tom's Guide, MakeUseOf). MakeUseOf Android Central
- **Claim mapping:** Razr Fold (2026) with its 6.6" cover and 8.1" inner is the most analogous to a "two-display body" envisioned by Claim 10. Literal "rear display unit" arguable: the cover IS on the back face of the folded device. DOE strong. MakeUseOf motorola
- **Severity: MEDIUM-HIGH (DOE) / LOW–MEDIUM (literal). Confidence: Medium.**

J. Microsoft Surface Duo 1 (Sept 2020) and Duo 2 (Oct 2021)

- **Display:** Two physically separate PixelSense OLED panels joined by a 360° hinge; explicitly NOT a single flexible display. Design Milk
- **Software:** Each screen can run a different Android app; spanned apps possible. Microsoft Support
- **Claim mapping:** Surface Duo arguably has a "front display unit" and "rear display unit" when folded 360°, but the patent emphasizes bending of a single flexible display, and Surface Duo's hinge motion is not "bending" in the patent's sense. Literal infringement very weak; DOE possible only under a very broad construction.
- **Severity: LOW. Confidence: Medium.**

K. LG Wing (Sept 2020) and LG G8X ThinQ

- LG Wing: Single main 6.8" plus a smaller 3.9" secondary OLED that SWIVELS out — not foldable, not bent, no "front/rear" pair on a continuous panel. LG USA
- LG G8X ThinQ: Phone + clip-on second screen case; not a single foldable body.
- **Severity: NONE for both. Confidence: High.**

L. Lenovo ThinkPad X1 Fold (Gen 1 2020; Gen 2 16-inch 2022), ASUS Zenbook 17 Fold, HP Spectre Foldable PC

- **Display:** Single 13.3"–16.3" foldable OLED. Hall-effect sensors trigger screen rotation/resolution changes (per Lenovo Tech Today engineering interview: "All the magnets on the PC and the accessories had to be placed in north-south orientations… They had to trigger a hall effect sensor to tell the software to switch the screen resolution and orientation or turn off the lower half of the screen when the keyboard is placed on top."). Lenovo Lenovo

- **Claim mapping:** When folded into "laptop" or "book" posture, one half is roughly flat while the other is angled — meeting "flat view area" and "bent view area" determinations of Claim 1(c). Software-controlled partition meets dependent claims 2, 5, 6.
- **Severity: MEDIUM. Confidence: Medium.** Caveat: the patent targets smartphones/tablets but "foldable electronic device" is not form-factor-limited.

M. Rollable concepts: Oppo X 2021, LG Rollable, TCL rollable, Motorola Rizr concept

- **Display:** Single OLED extending via roll motor (Oppo X 2021: 6.7"→7.4"). Engadget
- **Claim mapping:** The patent expressly covers rolling (dep. Claim 3 references rolling; dep. claims 7 [squeeze] and 8–9 [roll] uniquely fit rollables). But rollables generally do NOT present a "rear display unit" — the rolled-up portion is inside the body, not on the rear face — so independent claims 1 and 10 are problematic.
- **Severity: LOW (independent) / MEDIUM (dep. 7–9 in concept). Confidence: Low-Medium.** Most never shipped commercially.

N. ZTE Axon M (2017)

- Two separate displays joined by 180° hinge; predecessor to Surface Duo. Same analysis.
- **Severity: LOW. Confidence: Medium.**

Summary Comparison Table

Product	Display Type	Front+Rear of Single Display?	Diff. Apps Front/Rear When Folded?	Literal Sev.	DOE Sev.	Confidence
Huawei Mate X / Xs / Xs 2	Outward wraparound OLED	YES	YES	**HIGH**	HIGH	High
Huawei Mate XT (tri-fold)	Multi-fold flexible OLED	YES (outward section)	YES (dual-app mode)	**HIGH**	HIGH	High
Royole FlexPai 1 / 2	Outward wraparound OLED	YES	YES (XDA confirmed)	**HIGH**	HIGH	Very High
Samsung Z Fold/Flip/TriFold	Inner + separate cover	NO	YES (App Continuity)	N/A (Samsung)	N/A	—

Product	Display Type	Front+Rear of Single Display?	Diff. Apps Front/Rear When Folded?	Literal Sev.	DOE Sev.	Confidence
Google Pixel Fold / 9 Pro Fold / 10 Pro Fold	Inner + separate cover	NO	YES (A14 QPR1+)	LOW	MEDIUM	Medium
OnePlus Open / Oppo Find N3 / N5	Inner + separate cover	NO	YES (swipe-up)	LOW	MEDIUM	Medium
Honor Magic V3 / V5	Inner + separate cover	NO	YES (esp. F2F Translation)	LOW	MED-HIGH	Medium
Xiaomi Mix Fold 3/4, Mix Flip	Inner + separate cover	NO	YES (Hover Mode)	LOW	MEDIUM	Medium
Motorola Razr Plus / Fold	Inner + large external cover	NO	YES (full apps on cover)	LOW-MED	MED-HIGH	Medium
Microsoft Surface Duo 1 / 2	Two separate panels + hinge	NO	YES	LOW	LOW	Medium
LG Wing	Main + swivel secondary	NO	YES	NONE	NONE	High
Lenovo X1 Fold / ASUS Zenbook Fold / HP Spectre Foldable	Single foldable OLED	Bent vs flat regions	YES (per half)	MEDIUM	MEDIUM	Medium
Oppo X 2021 / LG Rollable (concepts)	Rollable OLED	NO rear face	N/A (single front)	LOW	LOW	Low-Med
ZTE Axon M	Two separate panels	NO	YES	LOW	LOW	Medium

Forward Citation Analysis

The Google Patents and Lens.org "Cited By" pages for US 9,354,742 B2 were not retrievable in this session. Adjacent Samsung patents in the same family of foldable-UI inventions include:

- **US 9,239,647 B2** — "Electronic device and method for changing an object according to a bending state" (Samsung; Korean priority 2012-08-20).
- **US 9,299,314 B2** — "Display device and method for controlling display image" (Samsung; Korean priority 2012-10-29).
- **US 11,372,446**, **US 11,429,149**, **US 12,014,102** — later Samsung foldable-UI patents likely citing US 9,354,742.

This is an acknowledged research gap; pulling the Image File Wrapper for U.S. Application No. 14/249,742 from USPTO Patent Center will close it.

Validity Considerations

- Filing date April 10, 2014 with Korean priority August 22, 2013. Prior art universe through mid-2013 applies.
- Earlier Samsung filings (US 9,239,647, US 9,299,314), Kyocera Echo (2011) dual-screen prior art, and pre-2014 Royole / Samsung Youm flexible-display demos form a dense field.
- The novelty hinges on combining (a) a single foldable display, (b) bend detection, and (c) UI on the bent rear area running a DIFFERENT application than the front. Pure dual-screen prior art lacks the "single foldable display" predicate; pure flexible-display prior art lacks the differentiated-application UI. A § 103 obviousness combination challenge would be credible but not trivial.

Recommendations

Stage 1 — immediate, low-cost: Treat Huawei Mate X family, Huawei Mate XT, and Royole FlexPai 1/2 as HIGH-priority literal-infringement targets. Practical enforcement is constrained by US sanctions on Huawei and Royole's bankruptcy, but a US ITC § 337 complaint against Huawei imports of Mate X devices, or a Chinese-court enforcement action, would be technically supportable. **Threshold to escalate:** confirm Mate X family US import volumes; document any pre-2024 Royole FlexPai 2 US developer sales.

Stage 2 — selective enforcement against book-style competitors: Against Pixel Fold, OnePlus Open, Honor Magic V3, Xiaomi Mix Fold, and Motorola Razr Fold, pursue doctrine-of-equivalents theories ONLY in conjunction with other Samsung foldable patents (Samsung holds hundreds), because standalone DOE assertion of US 9,354,742 is contestable. The patent's "curved visible region" specification language gives defendants a strong literal-claim-construction argument against separate-cover-screen products. **Threshold to escalate:** a Markman ruling in any related Samsung foldable case construing "rear display unit" broadly enough to include a separate physical cover panel.

Stage 3 — defensive monitoring: Pull the Image File Wrapper for App. 14/249,742 from USPTO Patent Center to confirm examiner-cited prior art and build a forward-citation map. Set up a quarterly PTAB monitor naming US 9,354,742 (none today, but tri-fold competitors may seek IPR if the patent is asserted).

Stage 4 — licensing posture: The most commercially valuable use of this patent is likely as a defensive cross-license counter-asset against Huawei and Honor's own foldable portfolios, rather than as an offensive monetization vehicle.

Caveats

- **Claim-construction risk is the single biggest variable.** The specification's "curved visible region" / "flat visible region" language and references to "folding and/or rolling" of a single foldable display strongly suggest the patentee envisioned a continuous flexible OLED (Mate X / FlexPai model), not a hinged two-panel device with a separate cover display. A court could construe "rear display unit" to require either (a) a region of a single continuous foldable display facing rearward when folded, or (b) any display physically located on the rear face of the device. The outcome materially shifts HIGH/LOW severity ratings for book-style foldables.
- **No litigation precedent.** Because US 9,354,742 has never been asserted, no court has construed its terms; all infringement analyses are predictive.
- **Forward-citation list is incomplete in this report.** Google Patents and Lens.org "Cited By" pages were not retrievable through the tools available; verification against the IFW is recommended before any pre-suit investigation.
- **Royole bankruptcy** (Shenzhen Intermediate People's Court accepted case May 15, 2024; liquidation began June 12, 2024) and **Huawei US sanctions** materially constrain Samsung's practical enforcement options against the HIGH-severity targets. CTOL Digital Solutions DIGITIMES
- **Samsung cannot infringe its own patent**; Galaxy Z Fold/Flip/TriFold analyses are for claim-scope calibration only.
- **The April 2026 Lepton Computing v. Samsung lawsuit** (Case No. 2:26-cv-00338, E.D. Tex., filed April 23, 2026; nine asserted patents, oldest filed June 29, 2021 per allaboutlawyer.com's analysis of the complaint) is unrelated to US 9,354,742 — Lepton is plaintiff, not defendant. Ex Parte AI
- **Dependent claims 7–9 (squeeze and roll events)** have not been commercially implemented in shipping foldables; the rollable concepts (Oppo X 2021, LG Rollable) that come closest were never released to market.
- **Source-quality note:** Several conclusions rely on developer documentation, manufacturer support pages, and third-party reviews (GSMArena, Android Authority, XDA, Engadget, Tom's Guide, Wikipedia). For litigation purposes these would need to be supplemented by direct product testing and source-code review of the alleged-infringer Android/EMUI/MagicOS builds.